

A FAIRNESS QUESTION FOR TWO FAMILIES OF INVOLUTIONS ON LATIN SQUARES

SETTING

Let L be a Latin square of order n on the symbol set $[n]$, with rows and columns indexed by $[n]$. Two rows, 1 and 2, and two columns, $j \neq j'$, are distinguished throughout. The frame. Let $\pi = \pi_L$ be the permutation of the rows defined by

$$L(\pi(r), j') = L(r, j) \quad (r \in [n]),$$

i.e. $\pi(r)$ is the row in which column j' holds the symbol that column j holds in row r ; we call π the *frame* of L . It is a derangement, since $\pi(r) = r$ would force $L(r, j) = L(r, j')$. The *bit* of L is

$$\beta(L) = \mathbf{1}\{\text{rows 1 and 2 lie in a common cycle of } \pi_L\},$$

and we write C_1, C_2 for the cycles of π_L through rows 1 and 2 (so $\beta = 1$ iff $C_1 = C_2$).

Row-pair permutations. For rows $x \neq y$ let $\rho_{x,y}$ be the permutation of the columns defined by $L(y, \rho_{x,y}(q)) = L(x, q)$. On the columns of any cycle Q of $\rho_{x,y}$ the two rows carry the same set of symbols.

Two families of moves.

- (TRADE) For rows $x \neq y$ with $x, y \notin \{1, 2\}$, let Q be the cycle of $\rho_{x,y}$ containing j' , and exchange the entries of rows x and y in the columns of Q .
- (TURN) For a cycle c of π_L containing neither row 1 nor row 2, exchange the entries of columns j and j' in the rows of c .

Lemma 1. *Both moves produce Latin squares, are involutions, and leave rows 1 and 2 unchanged. A turn replaces π by π^{-1} on the rows of c and leaves it unchanged elsewhere; the cycle partition is unchanged. A trade at (x, y) replaces π by $(xy) \circ \pi$ if $j \notin Q$ (so the cycles of x and y are merged if distinct and split if equal), and by $(xy) \circ \pi \circ (xy)$ if $j \in Q$ (the cycle partition is unchanged up to exchanging the labels x, y). In particular a trade can change β only when $j \notin Q$; call such a pair (x, y) flippable. If $\pi(x) = y$ or $\pi(y) = x$ then (x, y) is not flippable.*

The proofs are direct verifications. Both rows carry the same symbols on Q , and both columns carry the same symbols on c , which gives the Latin property; the traded cycle is again the $\rho_{x,y}$ -cycle of j' in the new square, and the turned cycle is again a cycle of the new frame, which gives the involution property. The frame formulas follow from the definition of π , since only the entries of rows x, y in columns j, j' matter. For the last claim: with $b' = L(y, j')$ and q^* the column in which row x holds b' , one has $\rho_{x,y}(q^*) = j'$, so Q is the $\rho_{x,y}$ -cycle of q^* . If $\pi(x) = y$ then $b' = L(x, j)$, so $q^* = j \in Q$; if $\pi(y) = x$ then $L(x, j') = L(y, j)$, so $\rho_{x,y}(j') = j \in Q$.

COMPONENTS AND THE QUESTION

Fix a $2 \times n$ Latin rectangle R (the first two rows) and columns $j \neq j'$ such that

$$(1) \quad R(1, j) \neq R(2, j') \quad \text{and} \quad R(1, j') \neq R(2, j),$$

i.e. $\pi(1) \neq 2$ and $\pi(2) \neq 1$ for every extension of R . (Without (1) the question below is void: if $R(1, j) = R(2, j')$ then $\pi(1) = 2$, hence $\beta = 1$, for every extension of R .) Let $S(R)$ be the set of Latin squares whose first two rows are R . Each move is an involution of $S(R)$; let $\mathcal{K}(R)$ be the set of orbits (“components”) of the group they generate, and for a component K let

$$p(K) = \frac{|\{L \in K : \beta(L) = 1\}|}{|K|}.$$

We call the bit *fair* on K if $p(K)$ is close to $\frac{1}{2}$.

Question 1 (component fairness). *Is it true that, uniformly over R and over column pairs (j, j') satisfying (1),*

$$\frac{1}{|S(R)|} \sum_{K \in \mathcal{K}(R)} |K| (2p(K) - 1)^2 \rightarrow 0 \quad (n \rightarrow \infty)?$$

Equivalently: for L uniform in $S(R)$, the conditional probability that $\beta(L) = 1$ given the component of L is $\frac{1}{2} + o(1)$ in L^2 . A quantitative version asks for the left side to be $o(1/\log^2 n)$. A weaker, averaged version that is still useful is stated in Remark 3(i).

Remark 1 (why the question is natural). Consider the Markov chain on $S(R)$ which, with probability $\frac{1}{2}$, picks a uniform pair $x, y \notin \{1, 2\}$ and trades, and otherwise picks a uniform row $x \notin \{1, 2\}$ and turns its π -cycle if that cycle avoids rows 1, 2 (holding if not). Each move is an involution chosen with state-independent probability, so the kernel is symmetric and the uniform measure on each component is stationary. Every state has a hold: by (1) the cycle C_1 is not $(1\ 2)$, so it contains a row $x \notin \{1, 2\}$ whose turn is rejected. Hence the chain is aperiodic on each component, and for the stationary chain started from the uniform measure on $S(R)$ the left side of the Question equals $\lim_{t \rightarrow \infty} \mathbb{E}[\beta'(L_0)\beta'(L_t)]$ with $\beta' = 2\beta - 1$, the long-time autocorrelation of the bit.

Along the chain the frame π performs the walk of left multiplication by transpositions (xy) , $x, y \notin \{1, 2\}$, *thinned* to the flippable pairs: the turns reverse a cycle and the non-flippable trades conjugate by (xy) , and neither changes β . The unthinned walk preserves $a = \pi^{-1}(1)$ and $b = \pi^{-1}(2)$. On the fibre $\{\pi : \pi^{-1}(1) = a, \pi^{-1}(2) = b\}$, identified with the permutations of $[n] \setminus \{1, 2\}$ by contracting the arcs $a \rightarrow 1 \rightarrow \pi(1)$ and $b \rightarrow 2 \rightarrow \pi(2)$, it is the walk by uniformly random transpositions, and rows 1, 2 share a cycle of π iff a, b share a cycle of the contracted permutation, which is the case for exactly half of the permutations of each fibre (two given points of a uniform permutation lie in a common cycle with probability exactly $\frac{1}{2}$). So if every pair were flippable the answer would be yes, with the left side $o(1)$ after $O(n \log n)$ steps by Diaconis–Shahshahani. The content of the Question is that the thinning by flippability does not destroy this.

Remark 2 (what is known numerically). The chain of Remark 1, modified to reject non-flippable trades (its components are finer than those above, so the evidence transfers), was run on uniformly random Latin squares of order $n = 30, 50, 100$, with (j, j') at distance at least 2 in both directions inside a cycle of the column permutation σ of R defined by $R(2, \sigma(q)) = R(1, q)$. A uniformly random pair $x, y \notin \{1, 2\}$ that is not π -adjacent (i.e. $\pi(x) \neq y, \pi(y) \neq x$) is flippable with probability 0.500 ± 0.001 , the same whether x, y lie in

one cycle of π or in two; the π -adjacent pairs, which are never flippable, are a fraction $\approx 2/n$ of all pairs. The autocorrelation of the bit, as a function of the number of trades performed, decays at least as fast as that of the unthinned walk and reaches the noise floor (≈ 0.01) within 64 trades at each n . Column pairs with j, j' in different cycles of σ were not tested at these sizes. Components have been enumerated only for $n \leq 6$, where the left side of the Question takes values between 0 (all tested pairs with j, j' in different σ -cycles) and 0.14.

Remark 3 (weaker forms that are still useful). (i) The application in Remark 4 needs only an averaged version: for every cycle type λ of σ , with L uniform among the Latin squares of order n whose column permutation σ has cycle type λ , R its first two rows, and $(j, j') = (j, \sigma^\alpha(j))$ with j in an m -cycle of σ and $\alpha, m - \alpha \geq 2$ prescribed, the expectation of the left side of the Question is $o(1/\log^2 n)$. (ii) A natural route is to show that, for all but an $o(1/\log^2 n)$ -fraction of $L \in S(R)$, a positive proportion of the bit-changing pairs are flippable — when $\beta = 1$ these are the pairs $x, y \in C_1$ such that rows 1 and 2 lie on different arcs of the split of C_1 into $(x, \pi(x), \dots, \pi^{-1}(y))$ and $(y, \pi(y), \dots, \pi^{-1}(x))$; when $\beta = 0$ they are the pairs $x \in C_1, y \in C_2$ — and to combine this with a mixing estimate for the thinned walk.

Remark 4 (provenance). The Question arose in a programme on the Cavenagh–Greenhill–Wanless conjecture on the cycle type of the second row of a random Latin square. Let $\mathcal{C}_n(\nu)$ be the number of Latin squares of order n whose column permutation σ (defined in Remark 2) has cycle type ν , let λ contain a part $m = \alpha + \alpha'$ with $\alpha, \alpha' \geq 2$, and let μ be λ with that part split into α, α' . For L uniform among the squares of type λ and (j, j') a uniform pair $\{j, \sigma^\alpha(j)\}$ with j in an m -cycle of σ , the marked-pair identity

$$2 \frac{\mathcal{C}_n(\lambda)}{\mathcal{C}_n(\mu)} - 1 = \frac{\mathbb{P}[\beta = 1]}{\mathbb{P}[\beta = 0]}$$

reduces the conjecture to $\mathbb{P}[\beta = 1] = \frac{1}{2} + o(1/\log n)$ over such instances, and a positive answer to the quantitative Question, averaged as in Remark 3(i), gives exactly that. The question itself makes no reference to this.